

To properly analyze Yousef and Lora's statements, we must first define what the habitable zone is. Simply stated, it is the orbital distance around a star within which the "temperature is right" for water to collect on the surface of a rocky planet. It is important to note that not only is the habitable zone a very narrow orbital region, but its location is wholly dependent on the spectral type of the host star. Massive, hot stars are more energetic and have their habitable zones further away, while small, cool stars are less energetic and are very close in. I will now utilize data from the Reference Table to demonstrate how both Yousef and Lora's statements are incorrect.

I disagree with Yousef's assumption that all three planets will possess liquid surface water based on the fact that they are all orbiting at 1 Astronomical Unit (AU). Yousef assumes that 1AU is an ideal, universal habitable distance because it is what the Earth has, and while true for Earth, according to the reference table, it is only within the habitable zone for a G-type star (0.9-1.8 AU). Exoplanet-1, which orbits an intensely hot A-type star, is well outside its habitable zone of 3.5-7.1 AU. At 1 AU, Exoplanet-1 would experience extremely high temperatures, thus evaporating all its surface water. For Exoplanet-3, orbiting an M-type star, the habitable zone is pulled much closer to the star, between 0.3 and 0.4 AU, thus making Exoplanet-3 freeze and only receive much less stellar radiation.

I disagree with Lora's assessment as well, but for partially different reasons than she suggested. Lora correctly assumes that Exoplanet-1 has no liquid water since it is in orbit around a massive and hot A-type star; an orbit of 1 AU is significantly within the "safe zone" of 3.5 to 7.1 AU for the A star and would boil off all surface water. Lora is also correct in her assessment of Exoplanet-2 because it is orbiting the G-type star at an appropriate distance of 1 AU which falls within its 0.9 to 1.8 AU habitable zone.

However, Lora is completely incorrect when it comes to Exoplanet-3. She assumed since the M-type star was listed as 'favorable' when looking for exoplanets, then any planet within its orbital proximity will have liquid water. She failed to consider the distance from the star. Because M-type stars are generally cooler and much less massive than other stars (0.08 to 0.52 solar masses), its habitable zone lies much closer, in between 0.3 to 0.4 AU. Since Exoplanet-3 orbits at a distance of 1 AU, it

is way beyond the outer limit of this habitable zone. This makes it a frozen ice world.

To conclude, among the three planets orbiting the G-type star at 1 AU will have the necessary orbital distance to support liquid surface water.